AQUAGRID AI / CS3081 / MILESTONE 01

A working multi-agent water-network prototype

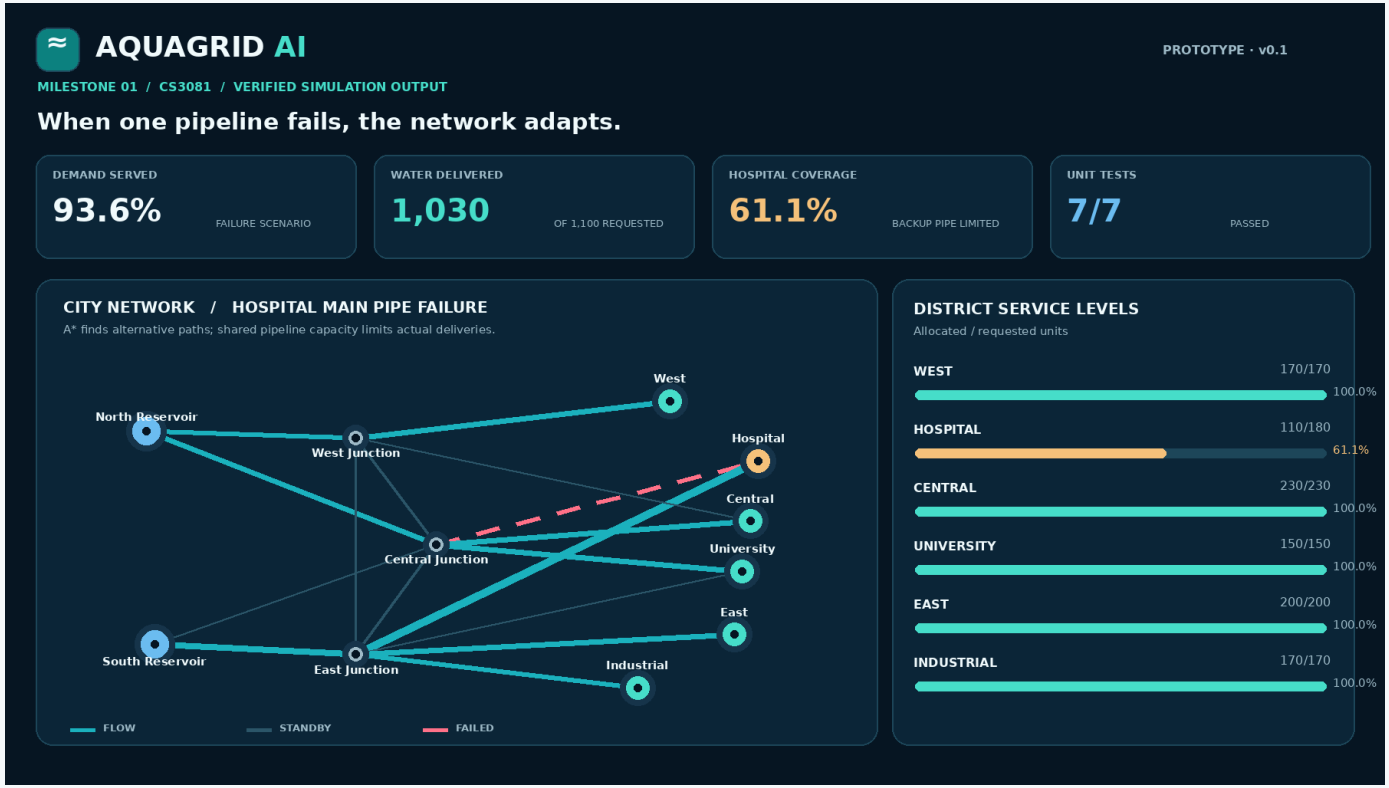
Prepared November 8, 2026 | Educational simulation using synthetic city data

7 / 7
Tests passed

5
Scenarios

100%
Normal demand met

93.6%
Pipe-break demand met



Implemented so far

Synthetic graph: 2 reservoirs, 6 districts, 3 junctions and 16 capacity-limited pipes.
4 agent roles exchange requests, offers, route proposals and allocation decisions.
A* searches available routes after failures.
Hospital receives priority; other districts share resources using cyclic allocation.

Demonstration

1. Open AquaGrid_Demo.html offline.
2. Switch to Hospital Pipe Failure.
3. Inspect backup path and service limits.
4. Run 7 Python tests as evidence.

NEXT: implement a static allocation baseline, repeatable experiments and deeper agent intelligence.

Limits: one time step, no water pressure or sensor uncertainty. Prepared with AI assistance; verify and attribute team contributions.